The Causal Relationship Between Volunteer Work and Satisfaction with Life

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PSYC300: Research Methods in Psychology

Section 0303

October 13, 2023

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The concept of one's mental well-being is a combination of many mental factors and contributors, one of which is satisfaction with life (SWL). Satisfaction with life refers to an individual's cognitive comparison of their current life position and what they deem to be the ideal standard (Diener et al., 1985). An individual with a larger SWL is said to possess a high appreciation for life's positives and negatives, an optimistic outlook on one's future and others, and a natural gratitude towards being alive and well every day (Veenhoven, 1996). These qualities highlight a reality that all humans wish to achieve, and as a result, highlight the importance of examining the numerous causes of life satisfaction.

Causal Variable

The causes of life satisfaction can range from emotional states regarding family, work, or school to life experiences such as social efforts, community engagement, or spending time with friends and family. Volunteer work refers to any type of behavior/engagement that betters individuals and/or a community, that is done without financial incentive (Van Willigen, 2000). Volunteering commonly allows individuals to gain perspective on both their own lives and the world around them, thus impacting how an individual feels about their life accomplishments, and current situation financially and emotionally (Marchesano & Musella, 2020). Furthermore, volunteering often promotes a more grateful, appreciative, and gratitude-filled mindset and lifestyle (Van Willigen, 2000). These associations all uphold the belief that volunteer work may be a possible contributor to an individual's SWL.

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Furthermore, numerous research papers and journal articles already back up this assertion. For example, one research article measured volunteering to see if it impacted an elderly population's SWL. The experiment involved a control group, who did not partake in volunteer work and an experimental group who did. Results found that elders participating in active unpaid social work had significantly higher life satisfaction and a stronger will to live in comparison to the control (Thoits, 2001). Many other studies have also found similar results using different populations or demographics. One study deduced that volunteer work is more positively correlated with SWL in elders as opposed to young adults (Van Willigen, 2000). Nevertheless, one study found no correlation between volunteer work and SWL in a national survey administered in Taiwan (Chen et al., 2014).

Current Study

Thus, the current association between volunteer work and SWL is relevant, but the direction of the correlation is still unknown. Therefore, these discrepancies confirm the need for further research on the correlation between volunteer work and SWL. The following research will aim to test whether or not there is a correlation between volunteer work and SWL. The hypothesis for the study is that there will be a positive correlation between volunteer work and SWL.

Proposed Methods

Population

The sample population for this study is undergraduate PSYC300 students at the University of Maryland. This is the population chosen as they are most willing to participate in the study. The subjects are predominantly female, with most living in or around College Park, Maryland at the time of the study. These participants were selected through convenience

sampling in which participants were the easiest to reach, the most willing, and available at the given time of the study. Furthermore, participants are given participation credit for using the university identification number to confirm their submission. All 13 respondents were viable and used in the correlational analysis.

Measures/Materials

Volunteer Work Scale

Volunteer work, the discrete predictor variable, was measured using a four-question Qualtrics survey. Some examples of questions include "I participate and engage with fundraising and non-profits" and "When I can, I give money/food to the homeless."Participants then rated their level of interaction with the statement using a 1-5 ordinally coded Likert scale, 1 meaning "never" and 5 meaning "always." After participants have responded to all four prompts, scores are ordinally summed using interval measurement. The higher the score the more active engagement the participant has with volunteer work. This scale system possesses high reliability through internal consistency seen through the coordinance of scores across all five statements (Price et al., 2015). However, the validity of this volunteer work scale is low to medium as this is the first time the specific scale is being used. Thus although there is some criterion validity, there is an extreme lack of discriminant validity that could be improved by employing this scale more frequently across larger and more world-representative sample sizes (see Appendix A).

Satisfaction with Life Scale

Satisfaction with life, the discrete outcome variable, was measured using the five-statement SWL scale (Diener et al., 1985). Some examples of statements include, " In most ways, my life is close to my ideal." and "If I could live my life over I would change nothing." Participants rated their agreement with statements on a 1-7 ordinally coded Likert scale, 1

meaning “ Strongly Disagree” and 7 meaning “ Strongly Agree” (Diener et al., 1985). After participants respond to all questions, scores are summed together using interval measurement. Higher aggregate scores indicate a greater SWL, while lower scores indicate a lower SWL. (Diener et al., 1985). This scale system possesses high reliability through internal consistency seen through the coordinance of scores across all five statements (Price et al., 2015). In regards to validity, the SWL scale has medium validity as although it contains criterion validity, in which the participant's scores are correlated with the variables they are expected to be coupled with, there is an evident need for discriminant validity to be established in future research (Diener et al., 1985) (see Appendix E).

Procedure

To gather data for this study, survey questions from the SWL scale and volunteer work scale were employed. Using Qulatrics software, the survey was constructed and distributed to participants (Qualtrics XM, 2023). After accessing the survey through an online link, participants were first asked to sign a virtual consent form, in which it is clear that they have the right to stop the survey at any time. Following this, participants will then complete a survey regarding both volunteer work and SWL. Lastly, participants will provide their UID number, to receive course credit, and will then be directed to a debriefing page where they will be informed of the purpose of the study and be thanked for their time. After collecting all of the data, it is cleaned using Microsoft Excel. All 13 participants fully responded to all portions of the survey, thus all information provided was utilized in Pearson’s R correlation analysis.

Design

A correlational design will be employed to measure if there is a relationship between volunteer work and SWL. This design is preferred as it limits the amount of excessive stress on

participants and does not depend on the manipulation of an artificial setting. For this design specifically, the predictor variable will be volunteer work and the criterion variable is SWL

Results

The demographic for this correlational analysis is comprised of undergraduate PSYC300 students at the University of Maryland. This convenience sample is mostly females who lived in College Park, Maryland at the time of the study. This test examined the hypothesis that there will be a positive correlation between volunteer work and SWL. The predictor variable being studied is engagement with volunteer work. The mean score for engagement with volunteer work was 10.46 with a standard deviation of 2.99. The outcome variable being studied is SWL. The mean score for SWL was 22.23 with a standard deviation of 8.07 (see Figure 1).

A Pearson r correlation coefficient was used to analyze the relationship between volunteer work and SWL. This is the most appropriate correlation coefficient as the summing of the data for both the predictor and outcome variables fall into the parameters of Pearson r in which both variables are interval/ratio based. The data in this study does not meet all four assumptions for Pearson r; the data is not normally distributed, the data is not linear, and the data does contain outliers. The assumptions not being met may heavily influence the validity of the correlation coefficient and may impact the interpretation of the data. According to the correlation results, there is a weak positive correlation between the variables with a r value of 0.23. However, the p-value for this correlation is 0.4358, thus the weak positive correlation depicted is not statistically significant (see Table 1). These results provide minimal support for the hypothesis that a positive correlation exists between volunteer work and SWL.

Discussion

For this correlational study, it was hypothesized that there is a positive correlation between volunteer work and SWL. The results depicted a weak positive correlation between the predictor and outcome variable, however, because the p-value is greater than 0.05, the original hypothesis is not supported and the results found are not statistically significant.

Strengths of Study

One strength of the research refers to the implementation of passive deception regarding the administration of the survey. Participants were not informed of the purpose of their survey involvement until after all answers were recorded. Thus, participants were more likely to answer both predictor and outcome questions honestly rather than trying to be a "good participant" and intentionally improve correlation results. Therefore, the ommittance of this information aided in the removal of participant bias, making the data recorded more reliable. Another strength of this research is the usage of a validated scale when collecting data for the outcome variable. The SWL scale is recognized as a highly reliable and valid scale employed across a multitude of studies (Diener et al., 1985). The implementation of this scale in this research study ensures that the information gathered for the outcome variable was done in a reputable way.

Limitations of Study

One weakness of this research is that the data does not meet some of the assumptions for a Pearson r correlational study. The data contains multiple outliers that can cause misleading results due to an inefficient line of best fit. Furthermore, the data is not normally distributed, thus the usage of a Pearsons r correlation coefficient is not the best choice as the chances of a type 1 error (falsely rejecting the null hypothesis) occurring are inflated and the reliability of the results is lower than it would be with a different correlation coefficient. Lastly, the lack of linear data makes the results unreliable and, much like the effect of not-normally distributed data, increases

the likelihood of a type 1 error occurring. Another weakness of this study is the convenience sampling method. The participants for this study are all PSYC300 students at the University of Maryland, with most being female and living in College Park, Maryland at the time of the study. The lack of participant diversity does not accurately represent the real-world population thus making the results less generalizable and lowering external validity.

Future Directions

Future directions for this correlational study include redoing the test with a larger population sample and using Spearman's rho correlation coefficient. In regards to improving the population sample, gathering a larger more diverse population by offering the study to the entirety of undergraduate PSYC students at the University of Maryland, or other universities, as opposed to just PSYC300, could provide results that hold more external validity than those conducted initially. Additionally, redoing this study with the improved population sample and using Spearman's rho instead of Pearson's r could produce more reliable and generalizable data as well. Spearman's rho is more reliable when using not normally distributed data and may produce more statistically significant results.

Concluding Thoughts

In regard to the overall impact on society, the urge to improve society's SWL is imperative and has been an active movement alongside many mental health initiatives. Therefore, this study may aid in figuring out what exactly makes an individual more satisfied with their lives, and one of those factors could be engaging with volunteer work more frequently.

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Appendix A

Volunteer Work Scale

Please rate your engagement with the following statements, with "Never" being on the far left and "Always" being on the far right. Please be honest and thoughtful with your response.

	Never	Rarely	Somtimes	Often	Always
When I see garbage/litter in public I make an effort to pick it up	○	○	○	○	○
I engage in community service opportunities	○	○	○	○	○
When I can, I give money/food to the homeless	○	○	○	○	○
I participate and engage with fundraising and non-profits	○	○	○	○	○

Appendix B

Satisfaction With Life Scale (Diener et al., 1985)

Below are five statements that you may agree or disagree with. Using the 1-7 scale below, indicate your agreement with each item by placing the appropriate number on the line preceding that item. Please be open and honest in your responding.

	Strongly disagree	Disagree	Slightly disagree	Neither agree nor disagree	Slightly agree	Agree	Strongly agree
In most ways, my life is close to ideal.	○	○	○	○	○	○	○
The conditions of my life are excellent.	○	○	○	○	○	○	○
I am satisfied with my life.	○	○	○	○	○	○	○
So far I have gotten the important things I want in life.	○	○	○	○	○	○	○
If I could live my life over, I would change almost nothing.	○	○	○	○	○	○	○

Table 1

Correlation Analysis for Volunteer Work and Satisfaction with Life

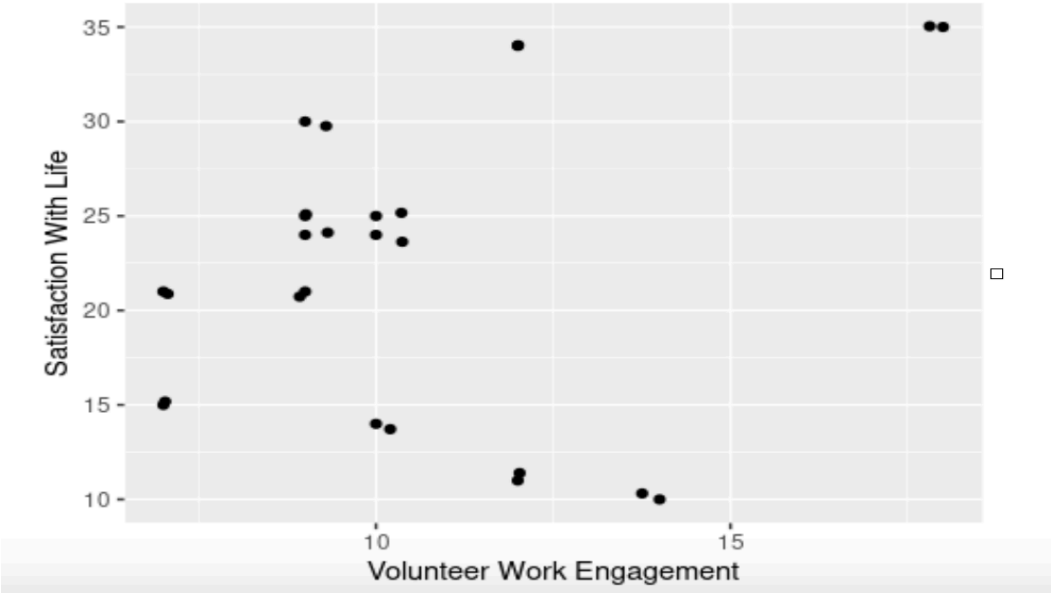
```
cor.test(predictor,outcome)

##
##  Pearson's product-moment correlation
##
## data:  predictor and outcome
## t = 0.80875, df = 11, p-value = 0.4358
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.3612330  0.6969201
## sample estimates:
##       cor
## 0.2369043
```

Note: this table depicts the correlation coefficient, p-value, and other data analysis for the sample.

Figure 1

Correlational Analysis of Volunteer Wok and Satisfaction with Life (Scatterplot)



Note: This figure represents the weak positive correlation between volunteer work (predictor variable on the y-axis) and satisfaction with life (outcome variable on the x-axis) that is not statistically significant, and thus does not support the original hypothesis.